

EXPERIMENT=2

TITLE $\Rightarrow$ Study and implementation of SQL fundamental DDL, DML Commands, Constraints, Data types, SELECT Queries, Filtering, Sorting, and Aggregate functions.

AIM $\Rightarrow$ To learn and implement SQL fundamentals including DDL and DML Commands. Constraints data types, SELECT queries with filtering and sorting and aggregate functions in a database.

RELATIONAL SCHEMA

Student

- $\Rightarrow$ Student ID (PK)
- $\Rightarrow$ Student Name
- $\Rightarrow$ Email (UNIQUE)
- $\Rightarrow$ Department ID (FK)

• DEPARTMENT

- $\Rightarrow$ Department ID (PK)
- $\Rightarrow$ Department Name.

COURSE

⇒ Course ID (PK)

⇒ Course Name

⇒ Credits

ENROLLMENT

⇒ Enroll ID (PK)

⇒ Student ID (FK)

⇒ Course ID (FK)

• Source Code (My SQL)

Create DATABASE CollegeDB;

Use CollegeDB;

~~CREATE TABLE Department (~~

~~Department ID INT Primary Key,~~

~~Department Name VARCHAR (50) Not Null Unique.~~
~~);~~

~~CREATE TABLE Student (~~

~~Student ID INT PRIMARY KEY,~~

~~Student Name VARCHAR (50) Not Null,~~

E-mail VARCHAR(100) Unique,

⇒ Department INT,

Foreign Key (Department ID)

References Department (Dept ID)

ON DELETE ~~Set~~ NULL

);

⇒ Create Table Course (

Course ID INT Primary Key,

Course Name Varchar(50) Not Null,

Credits INT ~~Not~~ NULL

);

⇒ CREATE TABLE Enrollment (

EnrollID INT Primary Key,

~~Student ID INT,~~

~~Course ID INT,~~

Foreign Key (student ID)

REFERENCES Student (Student ID)

ON Delete Cascade,

Foreign Key (Course ID)

References Course (Course ID)

On ~~DELETE~~ CASCADE

);

ent (Student ID)

Output

Department

Dept+ID

1

2

Dept Name

CSE

AI & ML

Student

Student ID

101

102

Student Name

Rahul

Priya

Email

rahul@gmail.com

priya@gmail.com

Dept ID

1

2

Course

Course ID

201

202

Course Name

DBMS

Operating System

Credits

4

4

Enrollment

Enroll ID

1

2

Student ID

101

102

Course ID

201

202

18/09/26

(Student ID)

ID)
ADE

Output

Department

Dept+ID

1

2

Dept Name

CSE

AI & ML

Student

StudentID

101

102

StudentName

Rahul

Priya

Email

rahul@gmail.com

~~Priya@gmail.com~~

DeptID

1

2

Course

CourseID

201

202

Course Name

DBMS

Operating System

Credits

4

4

Enrollment

Enroll ID

1

2

~~StudentID~~

~~101~~

102

CourseID

201

202

~~18/09/26~~